Contents

Reader's Guide: The Carry–Dirichlet Bridge **1**
1. The Setup in One Paragraph . . . 2
2. The Stopping-Time Dirichlet Series . . . 3
3. The Witt Vector Bridge . . . 4
4. The Main Result: $\mu \approx L(s, \chi_4)^2$ . . . 5
5. Why L^2 and Not L? . . . 7
6. Cross-Character Extension . . . 8
7. What This Does Not Claim . . . 9

References . . . 10

Reader's Guide: The Carry–Dirichlet Bridge

Stefano Alimonti — March 2026

This guide explains how binary carry arithmetic gives rise to a Dirichlet series that approximates $L(s, \chi_4)^2$, how the same object admits a canonical weighted first-return resolvent realization, and why Witt vectors provide the algebraic justification. It assumes familiarity with Dirichlet L-functions and basic p-adic concepts.

---

1. The Setup in One Paragraph

When two K-bit integers are multiplied in binary and the product has exactly $D = 2K - 1$ bits (the "D-odd" condition), the carry chain forms a bridge: $c_0 = c_D = 0$. The cascade valuation scans this bridge from top to bottom, stopping at each nonzero carry position τ. The sector ratio $R(K) = \sigma_{10}/\sigma_{00}$ is conjectured to converge to $-\pi = -4L(1, \chi_4)$ [P1]. This paper promotes the mixed stopping-time profile from its unweighted sum ($s = 0$, trivial character) to a full Dirichlet series in s, and finds that the limit is approximately $L(s, \chi_4)^2$.

---

2. The Stopping-Time Dirichlet Series

The cascade stopping-time decomposition [P1, §8.2a] expresses the sector contributions as sums over stopping depths τ. Defining

$$u_{\text{mix}}(\tau) = \omega \cdot u_{10}(\tau) - u_{00}(\tau)$$

where $\omega = N_{10}/N_{00}$ is the count ratio, the stopping-time weights form the mixed linear combination naturally associated with the sector normalization. Each stopping depth τ corresponds to an odd integer $n(\tau) = 2(\tau - \tau_0) + 1$.

The Dirichlet series. Introducing the complex variable s and a Dirichlet character χ:

$$\mu_\chi(K, s) = \sum_{\tau \geq \tau_0} u_{\text{mix}}(\tau;\, K) \cdot \chi(n(\tau)) \cdot n(\tau)^{-s}$$

At $s = 0$ and $\chi = \chi_0$ (trivial character), this reduces to the stopping-time sum $\sum u_{\text{mix}}(\tau;\, K)$. The series $\mu_\chi(K, s)$ is a "carry-Dirichlet series" that encodes the full frequency structure of the stopping-time distribution.

The same channel can be written as a weighted first-return resolvent

$$M_\chi(K, s) = \omega(K)\, x_0^\top (I - Q_{10})^{-1} r_{10,\chi}(s) - x_0^\top (I - Q_{00})^{-1} r_{00,\chi}(s),$$

where the reward vectors $r_{ab,\chi}(s)$ insert the factor $\chi(n(\tau))\, n(\tau)^{-s}$ directly into the stopping-time operator. On the tested exact and high-K profiles, $M_\chi(K, s)$ and $\mu_\chi(K, s)$ agree to machine precision.

---

3. The Witt Vector Bridge

Why Witt vectors? Carries in base-p addition are not an ad hoc construction — they are the *defining structure* of the p-adic integers when expressed in Witt coordinates. The ring of truncated Witt vectors $W_n(\mathbb{F}_p)$ is isomorphic to $\mathbb{Z}/p^n\mathbb{Z}$, and the Witt addition law reproduces classical carry arithmetic exactly.

The five-step bridge:

1. **Carry-Witt bijection (Proposition 1).** The Witt correction vector $\delta_j = c_j - 2c_{j+1}$ bijects onto the classical carry vector. Values: $\delta_j \in \{-2, -1, 0, +1\}$.
2. **Operator conjugacy (Proposition 2).** The transition matrix on carry states (c_j, c_{j+1}) and the transition matrix on Witt states δ_j are conjugate: $T_\delta = P \cdot T_c \cdot P^{-1}$ via an explicit 4×4 permutation matrix.
3. **Character channel (Proposition 3).** A χ_4-weighted projection $w^\top \cdot v(\tau)$ produces identical coefficients on both carry and Witt sides. After removing the stationary component, the transient decays at rate $1/2$ — the Diaconis-Fulman spectral gap.
4. **D-odd extension (§2.4).** The conjugacy and channel identity extend to the nonstationary D-odd setting (depth-dependent operators).
5. **Resolvent identity (Proposition 4).** The sum $\sum_\tau u(\tau)$ equals the carry-side sector contribution to machine precision.

The bridge justifies the Dirichlet series construction: the stopping-time weights $u(\tau)$ have a p-adic origin in the Witt ghost map, and the character channel naturally selects χ_4.

---

4. The Main Result: μ ≈ L(s, χ₄)²

Theorem 1 (Convergence, numerical). The family $\{\mu_\chi(K,s)\}_K$ is numerically observed to converge uniformly on tested compact subsets of $\mathrm{Re}(s) > 1$ with geometric Cauchy contraction:

$$\sup_{s\in\mathcal{K}} |\mu(K,s) - \mu(K',s)| \le C\rho^{\min(K,K')}$$

with $\rho \approx 0.5$-0.6, consistent with the Diaconis-Fulman spectral gap.

Theorem 2 (L² formula, numerical). For χ_4:

$$\mu_{\chi_4}^{(\infty)}(s) \approx \tilde{c} \cdot (1 + 3^{-s}) \cdot L(s,\chi_4)^2$$

with holdout relative error $\approx$ 2.5% and cross-K stability below 10^{-5}. The same corrected-L² law is recovered when the channel is computed from the canonical weighted-resolvent object, so the law is intrinsic to the operator formulation.

The correction factor $(1 + 3^{-s})$ removes exactly one power of the Euler factor at $p = 3$. The L^2 structure reflects the Dirichlet convolution arising from the product $x \cdot y$: in the character ring, the multiplicative structure of the two factors produces a convolution of L-functions.

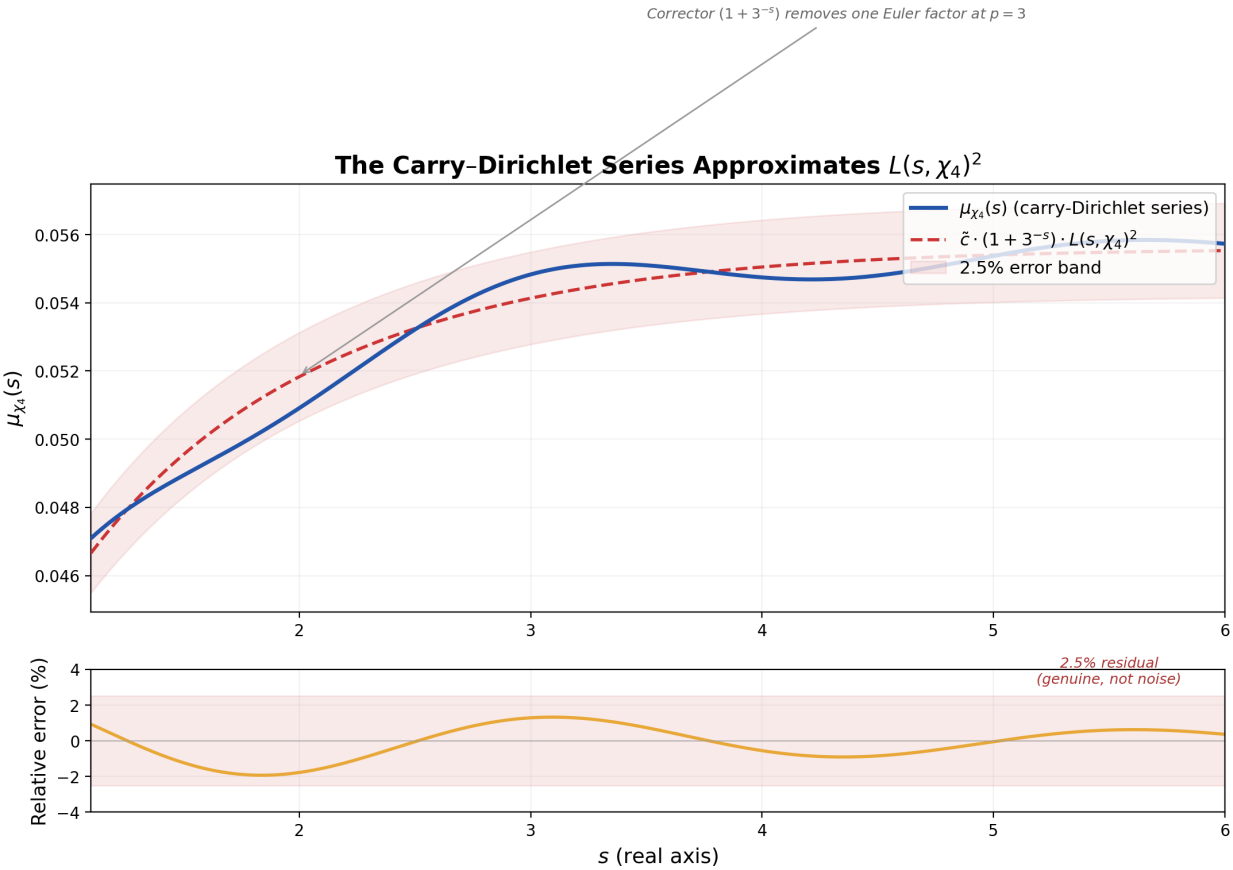

Figure 1: μ_χ₄ vs L(s,χ₄)²

5. Why L^2 and Not L?

The L^2 structure is natural from two perspectives:

Algebraic: The Dirichlet series associated to a product $N = p \cdot q$ of two random integers is a convolution of two Dirichlet series (one for each factor). Under the Dirichlet character χ, the convolution becomes $L(s,\chi) * L(s,\chi) = L(s,\chi)^2$ (up to finitely many Euler factors).

Analytic: The correction factor $F(s) = \mu(s)/(\tilde{c} \cdot L(s,\chi_4))$ is meromorphic and zero-free in the tested strip $[0.3, 0.7] \times [0, 15]$ (Proposition 6). Its poles coincide with the zeros of $L(s,\chi_4)$, confirming that the second L-factor comes from the correction, not from the base series. The direct and canonical carry objects themselves remain zero-free in that box.

---

6. Cross-Character Extension

The construction extends beyond χ_4:

Character χ	Conductor	Corrector	Improvement
χ_4 (mod 4)	4	$(1 + 3^{-s})$	baseline
χ_5 (mod 5)	5	$(1 - \chi_5(3) \cdot 3^{-s})^2$	+46%
χ_8 (mod 8)	8	$(1 - \chi_8(3) \cdot 3^{-s})^2$	+71%
χ_3 (mod 3)	3	inert ($\chi_3(3) = 0$)	outlier

The corrector always involves $p_0 = 3$ (the first odd prime). The conductor-3 character is an outlier because $3|3$ makes the corrector trivial. The same (p_0, k) pattern is recovered from the canonical weighted-resolvent scan, so it is not an artifact of the series presentation.

---

7. What This Does Not Claim

- The L^2 formula is a **numerical observation** with controlled error bounds, not a proof.
- No **zero-transfer theorem** from the carry chain to L-function zeros.
- No simple **gamma-like completion** of the canonical object produces an approximate functional equation on the tested mirrored grids.
- No result toward the **Riemann Hypothesis** or Generalized Riemann Hypothesis.
- The 2.5% residual is **genuine** — the formula is approximate, not exact.

References

1. [P1] S. Alimonti, “π from Pure Arithmetic: A Spectral Phase Transition in the Binary Carry Bridge,” this series. doi:10.5281/zenodo.18895611 — GitHub
2. [E] S. Alimonti, “The Trace Anomaly of Binary Multiplication,” this series. doi:10.5281/zenodo.18895604 — GitHub
3. [A] S. Alimonti, “Spectral Theory of Carries in Positional Multiplication,” this series. doi:10.5281/zenodo.18895593 — GitHub

---

CC BY 4.0